

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1434

Ladysmith 138 kV
January 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1434. The Interconnection Customer proposes a 260.3 MW battery energy storage facility interconnecting at the Ladysmith 138 kV substation in the Virginia area. The request was received on March 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 566 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$55,625,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dunn Siding to Ladysmith Substation 138 KV	Merced to Tuckasegee Tie 138 KV	159.4	244	81.8
Cranes Corner to Re River Substation 138 KV	Dunn Siding to Ladysmith 138 KV	155.9	204	100.0
Cranes Corner Substation to Calmont 138 KV L	Dunn Siding to Ladysmith 138 KV	135.8	217	100.0
Birchwood 345/138kV XFMR	Birchwood Substation to Nauvoo 345 KV	114.2	529	5.6
Calmont to Ladysmith 138 KV L	Ladysmith to Dunn Siding 138 KV	112.6	240	100.0
Slabtown to Dunn Siding 138 KV L	Libby to Red Riv 138 KV	112.5	271	13.3
Lookout to Nathan 138 KV	Libby to Contra Costa 138 KV	109.9	321	6.1
Alligator 500 kV/345 kV Autotransformer	Mountaineer Wind Energy Ctr to Santee Cooper 345 KV L	107.0	896	5.1
Westlake to Lake Agnes 138 KV line	Dunn Siding to Ladysmith 138 KV	102.9	228	5.1
Kearneyville to Clark County 345 KV	Grimmway-Malaga to Sleepy Hollow 138 KV	102.2	1,721	6.6
Sperry to Libby 138 KV Line	Red River to Libby 138 KV line	102.0	292	18.0
Mountaineer Wind Energy Center to Santee Cooper 345 KV L	Spindle Substation 500/345 kV Transformer	101.8	1,476	5.1
Skaggs Branch to Sperry 138 KV	Red Riv to Libby 138 KV line	101.1	329	18.0
Slabtown to Skaggs Branch Substation 138 KV	Red River to Libby 138 KV line	100.6	361	18.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dunn Siding to Ladysmith Substation 138 KV	Rebuild line with higher-capacity conductor	\$4,791,000	100% (\$4,791,000)
Cranes Corner to Red Riv 138 KV	Rebuild line with higher-capacity conductor	\$3,344,000	100% (\$3,344,000)
Cranes Corner Substation to Calmont 138 KV L	Rebuild line with higher-capacity conductor	\$5,764,000	100% (\$5,764,000)
Birchwod 345/138kV TX	Replace transformer with higher-rated unit	\$16,208,000	45% (\$7,231,000)
Calmnt to Ladysmith Substation 138 KV	Rebuild line with higher-capacity conductor	\$10,270,000	100% (\$10,270,000)
Slabtown to Dunn Siding 138 KV	Rebuild line with higher-capacity conductor	\$13,552,000	100% (\$13,552,000)
Nathan to Lookout 138 KV	Reconductor line	\$4,195,000	14% (\$595,000)
Alligator 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$32,894,000	3% (\$1,103,000)
Westlake to Lake Agnes 138 KV line	Reconductor line	\$595,000	58% (\$347,000)
Kearneyville to Clark County 345 KV	Reconductor line	\$25,737,000	2% (\$484,000)
Sperry to Libby 138 KV Line	Reconductor line	\$562,000	100% (\$562,000)
Mountaineer Wind Energy Center to Santee Cooper 345 KV L	Reconductor line	\$15,888,000	4% (\$644,000)
Skaggs Branch to Sperry 138 KV	Reconductor line	\$4,606,000	100% (\$4,606,000)
Slabtown to Skaggs Branch 138 KV	Reconductor line	\$2,332,000	100% (\$2,332,000)
	Total allocated to project		\$55,625,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.